ASSIGNMENT:- 1

AIM:- Frequency of words using MapReduce

PROBLEM STATEMENT:- Develop a MapReduce program to calculate the frequency of a given word in a given file.

THOERY:-

Introduction

In today's data-driven world, organizations generate and store massive volumes of data from various sources like social media, sensors, transactions, and logs. Analyzing such large-scale datasets using traditional single-machine systems is inefficient and often impossible due to limitations in storage and processing power.

To tackle these challenges, frameworks like Apache Hadoop were introduced. Hadoop allows for the *distributed storage* and **parallel processing** of large datasets across a cluster of commodity hardware.

One of the core components of Hadoop is MapReduce, a programming model and processing engine that enables parallel computation of data by dividing tasks into two fundamental phases: *Map* and *Reduce*.

This practical is a classic example of using the MapReduce model to perform a word count operation, where the program scans a large text file and counts the frequency of each word. It is often considered the "Hello World" of MapReduce because it demonstrates the power of distributed processing through a simple and understandable task.

Hadoop Streaming

While Hadoop is typically used with Java, Hadoop Streaming is a powerful utility that allows users to write Mapper and Reducer code in any language that can read from standard input and write to standard output, such as Python, Perl, Bash, or Ruby.

This enables greater flexibility and rapid development. In this practical, we use Python for its simplicity and readability.

Working Principle of Word Count using MapReduce

The process follows these four key steps:

1. Input File (HDFS):- The input to the MapReduce job is a plain text file stored in *Hadoop Distributed File System (HDFS)*. This file is split into blocks and distributed across the cluster for parallel processing.

2. Mapper Phase

The Mapper reads the input line by line. For each line, it performs the following:

- Splits the line into words based on whitespace and punctuation.

- For each word, emits a key-value pair in the form of:
  `(word, 1)`

Example
Input to Mapper:- *Hadoop is an open-source framework*
Output from Mapper:-

```
Hadoop    1
is   1
an   1
open-source    1
framework    1
```

- This is the intermediate output, which is not the final result

3. Shuffle and Sort (Handled by Hadoop Framework)
This is an internal and automatic phase managed by Hadoop.
- All intermediate key-value pairs are shuffled, meaning they are sent to the appropriate reducer based on the key.
- Hadoop then groups the values by key, which means it collects all values (1s) for the same word together.

```
Hadoop → [1, 1]
is → [1]
an → [1]
open-source → [1]
framework → [1]
```

4. Reducer Phase
The Reducer receives each unique key (word) and a list of its values (counts). It then:
- Sums up the values to get the total occurrences of each word.
- Emits the final key-value pair as:
  `(word, total_count)`

Output from Reducer:-

```
Hadoop    2
is   1
an   1
open-source    1
framework    1
```

- This is the final output, which shows how many times each word appeared in the input file.

Advantages of MapReduce in Word Count:
- Parallelism: The program runs across multiple nodes, increasing speed and efficiency.
- Fault Tolerance: If one node fails, Hadoop re-runs the task on another node.
- Scalability: Easily scalable by adding more nodes to the cluster.

- Simplicity: Abstracts the complexity of distributed systems behind a simple programming model.

CODE:-

Start Hadoop Services

1. Open terminal and switch to the Hadoop user

```
~$ su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd
```

2. Start HDFS

```
hduser@Ubuntu:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu]
2025-04-14 17:16:47,579 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

3. start-yarn.sh

```
hduser@Ubuntu:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers

hduser@Ubuntu:~$ jps
3891 Jps
3141 DataNode
3352 SecondaryNameNode
3000 NameNode
```

4. Create a text file with content to count words

```
# making wordcount.txt file
hduser@Ubuntu:~$ ls
Desktop    Downloads    hadoop-3.3.4.tar.gz  Pictures  snap       Videos
Documents  hadoop-3.3.4  Music                Public    Templates

hduser@Ubuntu:~$ nano word_count.txt
```

{

Wordcount.txt file will open in nano

}

```
hduser@Ubuntu:~$ cat word_count.txt
Hadoop is an open-source framework used for storing and processing large datasets
across clusters of computers. It uses a distributed file system (HDFS) and the
MapReduce programming model. Designed for scalability and fault tolerance, Hadoop
enables efficient data analysis and is widely used in big data applications across
industries.

hduser@Ubuntu:~$ hadoop fs -ls /
2025-04-14 17:33:13,851 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

5. Create a directory in HDFS and Upload your file

```
hduser@Ubuntu:~$ hadoop fs -mkdir /mapreduce_word_count
```

```
2025-04-14 17:33:36,097 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /

```
2025-04-14 17:33:39,481 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x   - hduser supergroup          0 2025-04-14 17:33 /
```

mapreduce_word_count

hduser@Ubuntu:~$ hadoop fs -put word_count.txt / mapreduce_word_count

```
2025-04-14 17:34:52,722 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /mapreduce_word_count

```
2025-04-14 17:35:33,599 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 hduser supergroup        338 2025-04-14 17:34 /
```

mapreduce_word_count/word_count.txt

hduser@Ubuntu:~$ hadoop fs -cat /mapreduce_word_count/ word_count.txt

```
2025-04-14 17:35:57,834 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

Hadoop is an open-source framework used for storing and processing large datasets across clusters of computers. It uses a distributed file system (HDFS) and the MapReduce programming model. Designed for scalability and fault tolerance, Hadoop enables efficient data analysis and is widely used in big data applications across industries.

6. Create Mapper and Reducer Python Scripts

hduser@Ubuntu:~$ nano mapper_word_count.py

{

After this command mapper_word_count.py will open in nano,
Write the mapper code for word_count

}

hduser@Ubuntu:~$ nano reducer_word_count.py

{

After this command reducer_word_count.py will open in nano,
Write the reducer code for word_count

}

hduser@Ubuntu:~$ ls

```
Desktop      hadoop-3.3.4        Music  reducer_word_count.py  Videos
Documents  hadoop-3.3.4.tar.gz   Pictures  snap                    word_count.txt
Downloads  mapper_word_count.py  Public Templates
```

hduser@Ubuntu:~$ chmod +x mapper_word_count.py

hduser@Ubuntu:~$ chmod +x reducer_word_count.py

7. Locate Hadoop Streaming JAR

Now open new terminal (keep the old terminal active) and note the path

su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd

hduser@Ubuntu:~$ cd /usr/local/hadoop/share/hadoop/tools/lib

hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ ls

```
aliyun-java-sdk-core-4.5.10.jar        hadoop-gridmix-3.3.4.jar
aliyun-java-sdk-kms-2.11.0.jar         hadoop-kafka-3.3.4.jar
aliyun-java-sdk-ram-3.1.0.jar          hadoop-minicluster-3.3.4.jar
aliyun-sdk-oss-3.13.0.jar              hadoop-openstack-3.3.4.jar
aws-java-sdk-bundle-1.12.262.jar       hadoop-resourceestimator-3.3.4.jar
azure-data-lake-store-sdk-2.3.9.jar    hadoop-rumen-3.3.4.jar
azure-keyvault-core-1.0.0.jar          hadoop-sls-3.3.4.jar
azure-storage-7.0.1.jar                hadoop-streaming-3.3.4.jar
hadoop-aliyun-3.3.4.jar                hamcrest-core-1.3.jar
hadoop-archive-logs-3.3.4.jar          ini4j-0.5.4.jar
hadoop-archives-3.3.4.jar              jdk.tools-1.8.jar
hadoop-aws-3.3.4.jar                   jdom2-2.0.6.jar
hadoop-azure-3.3.4.jar                 junit-4.13.2.jar
hadoop-azure-datalake-3.3.4.jar        kafka-clients-2.8.1.jar
hadoop-client-3.3.4.jar                lz4-java-1.7.1.jar
hadoop-datajoin-3.3.4.jar              ojalgo-43.0.jar
hadoop-distcp-3.3.4.jar                opentracing-api-0.33.0.jar
hadoop-dynamometer-blockgen-3.3.4.jar  opentracing-noop-0.33.0.jar
hadoop-dynamometer-infra-3.3.4.jar     opentracing-util-0.33.0.jar
hadoop-dynamometer-workload-3.3.4.jar  org.jacoco.agent-0.8.5-runtime.jar
hadoop-extras-3.3.4.jar                wildfly-openssl-1.0.7.Final.jar
hadoop-fs2img-3.3.4.jar                zstd-jni-1.4.9-1.jar
```

hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ pwd

/usr/local/hadoop/share/hadoop/tools/lib

8. Go back to your old terminal!
Paste the directory got from new terminal

/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar

hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -input /mapreduce_word_count/word_count.txt -output /output -mapper /home/hduser/mapper_word_count.py -reducer /home/hduser/reducer_word_count.py

**Make sure you correctly give your own saved txt and py file names.

```
2025-04-14 18:00:52,984 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2025-04-14 18:00:53,779 INFO impl.MetricsConfig: Loaded properties from hadoop-
metrics2.properties
2025-04-14 18:00:54,061 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period
at 10 second(s).
2025-04-14 18:00:54,061 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-14 18:00:54,097 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-14 18:00:54,444 INFO mapred.FileInputFormat: Total input files to process : 1
2025-04-14 18:00:54,743 INFO mapreduce.JobSubmitter: number of splits:1
```

```
2025-04-14 18:00:55,362 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_local1682281268_0001
2025-04-14 18:00:55,370 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-14 18:00:55,613 INFO mapreduce.Job: The url to track the job: http://
localhost:8080/
2025-04-14 18:00:55,624 INFO mapreduce.Job: Running job: job_local1682281268_0001
2025-04-14 18:00:55,627 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-04-14 18:00:55,642 INFO mapred.LocalJobRunner: OutputCommitter is
org.apache.hadoop.mapred.FileOutputCommitter
2025-04-14 18:00:55,658 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-14 18:00:55,658 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-14 18:00:55,775 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-04-14 18:00:55,780 INFO mapred.LocalJobRunner: Starting task:
attempt_local1682281268_0001_m_000000_0
2025-04-14 18:00:55,850 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-14 18:00:55,850 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-14 18:00:55,886 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-04-14 18:00:55,902 INFO mapred.MapTask: Processing split: hdfs://localhost:54310/
mapreduce_word_count/word_count.txt:0+338
2025-04-14 18:00:55,953 INFO mapred.MapTask: numReduceTasks: 1
2025-04-14 18:00:56,047 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-04-14 18:00:56,048 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-04-14 18:00:56,049 INFO mapred.MapTask: soft limit at 83886080
2025-04-14 18:00:56,049 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-04-14 18:00:56,049 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-04-14 18:00:56,063 INFO mapred.MapTask: Map output collector class =
org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-04-14 18:00:56,068 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
mapper_word_count.py]
2025-04-14 18:00:56,075 INFO Configuration.deprecation: mapred.work.output.dir is
deprecated. Instead, use mapreduce.task.output.dir
2025-04-14 18:00:56,077 INFO Configuration.deprecation: mapred.local.dir is
deprecated. Instead, use mapreduce.cluster.local.dir
2025-04-14 18:00:56,084 INFO Configuration.deprecation: map.input.file is deprecated.
Instead, use mapreduce.map.input.file
2025-04-14 18:00:56,084 INFO Configuration.deprecation: map.input.length is
deprecated. Instead, use mapreduce.map.input.length
2025-04-14 18:00:56,084 INFO Configuration.deprecation: mapred.job.id is deprecated.
Instead, use mapreduce.job.id
2025-04-14 18:00:56,085 INFO Configuration.deprecation: mapred.task.partition is
deprecated. Instead, use mapreduce.task.partition
2025-04-14 18:00:56,090 INFO Configuration.deprecation: map.input.start is deprecated.
Instead, use mapreduce.map.input.start
2025-04-14 18:00:56,090 INFO Configuration.deprecation: mapred.task.is.map is
deprecated. Instead, use mapreduce.task.ismap
2025-04-14 18:00:56,090 INFO Configuration.deprecation: mapred.task.id is deprecated.
Instead, use mapreduce.task.attempt.id
2025-04-14 18:00:56,090 INFO Configuration.deprecation: mapred.tip.id is deprecated.
Instead, use mapreduce.task.id
2025-04-14 18:00:56,091 INFO Configuration.deprecation: mapred.skip.on is deprecated.
Instead, use mapreduce.job.skiprecords
2025-04-14 18:00:56,091 INFO Configuration.deprecation: user.name is deprecated.
Instead, use mapreduce.job.user.name
2025-04-14 18:00:56,328 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-14 18:00:56,333 INFO streaming.PipeMapRed: Records R/W=1/1
2025-04-14 18:00:56,336 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-14 18:00:56,339 INFO streaming.PipeMapRed: mapRedFinished
2025-04-14 18:00:56,350 INFO mapred.LocalJobRunner:
2025-04-14 18:00:56,350 INFO mapred.MapTask: Starting flush of map output
2025-04-14 18:00:56,350 INFO mapred.MapTask: Spilling map output
2025-04-14 18:00:56,350 INFO mapred.MapTask: bufstart = 0; bufend = 436; bufvoid =
104857600
2025-04-14 18:00:56,350 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend =
26214204(104856816); length = 193/6553600
2025-04-14 18:00:56,410 INFO mapred.MapTask: Finished spill 0
```

2025-04-14 18:00:56,466 INFO mapred.Task: Task:attempt_local1682281268_0001_m_000000_0 is done. And is in the process of committing
2025-04-14 18:00:56,487 INFO mapred.LocalJobRunner: Records R/W=1/1
2025-04-14 18:00:56,487 INFO mapred.Task: Task 'attempt_local1682281268_0001_m_000000_0' done.
2025-04-14 18:00:56,509 INFO mapred.Task: Final Counters for attempt_local1682281268_0001_m_000000_0: **Counters: 23**

File System Counters
FILE: Number of bytes read=141405
FILE: Number of bytes written=787271
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=338
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=1
Map output records=49
Map output bytes=436
Map output materialized bytes=540
Input split bytes=110
Combine input records=0
Spilled Records=49
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=168820736
File Input Format Counters
Bytes Read=338

2025-04-14 18:00:56,511 INFO mapred.LocalJobRunner: Finishing task: attempt_local1682281268_0001_m_000000_0
2025-04-14 18:00:56,512 INFO mapred.LocalJobRunner: map task executor complete.
2025-04-14 18:00:56,532 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2025-04-14 18:00:56,533 INFO mapred.LocalJobRunner: Starting task: attempt_local1682281268_0001_r_000000_0
2025-04-14 18:00:56,569 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-04-14 18:00:56,573 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-04-14 18:00:56,574 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2025-04-14 18:00:56,589 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@400253b0
2025-04-14 18:00:56,600 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2025-04-14 18:00:56,645 INFO mapreduce.Job: Job job_local1682281268_0001 running in uber mode : false
2025-04-14 18:00:56,652 INFO mapreduce.Job: map 100% reduce 0%
2025-04-14 18:00:56,674 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=546098368, maxSingleShuffleLimit=136524592, mergeThreshold=360424928, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2025-04-14 18:00:56,679 INFO reduce.EventFetcher: attempt_local1682281268_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2025-04-14 18:00:56,777 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1682281268_0001_m_000000_0 decomp: 536 len: 540 to MEMORY
2025-04-14 18:00:56,780 INFO reduce.InMemoryMapOutput: Read 536 bytes from map-output for attempt_local1682281268_0001_m_000000_0

```
2025-04-14 18:00:56,792 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output
of size: 536, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->536
2025-04-14 18:00:56,806 INFO reduce.EventFetcher: EventFetcher is interrupted..
Returning
2025-04-14 18:00:56,809 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-14 18:00:56,811 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-
memory map-outputs and 0 on-disk map-outputs
2025-04-14 18:00:56,838 INFO mapred.Merger: Merging 1 sorted segments
2025-04-14 18:00:56,839 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 527 bytes
2025-04-14 18:00:56,850 INFO reduce.MergeManagerImpl: Merged 1 segments, 536 bytes to
disk to satisfy reduce memory limit
2025-04-14 18:00:56,850 INFO reduce.MergeManagerImpl: Merging 1 files, 540 bytes from
disk
2025-04-14 18:00:56,850 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from
memory into reduce
2025-04-14 18:00:56,851 INFO mapred.Merger: Merging 1 sorted segments
2025-04-14 18:00:56,851 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 527 bytes
2025-04-14 18:00:56,853 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-14 18:00:56,855 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
reducer_word_count.py]
2025-04-14 18:00:56,866 INFO Configuration.deprecation: mapred.job.tracker is
deprecated. Instead, use mapreduce.jobtracker.address
2025-04-14 18:00:56,873 INFO Configuration.deprecation: mapred.map.tasks is
deprecated. Instead, use mapreduce.job.maps
2025-04-14 18:00:57,005 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-14 18:00:57,007 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-14 18:00:57,014 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-14 18:00:57,015 INFO streaming.PipeMapRed: Records R/W=49/1
2025-04-14 18:00:57,021 INFO streaming.PipeMapRed: mapRedFinished
2025-04-14 18:00:57,134 INFO mapred.Task: Task:attempt_local1682281268_0001_r_000000_0
is done. And is in the process of committing
2025-04-14 18:00:57,144 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-14 18:00:57,144 INFO mapred.Task: Task attempt_local1682281268_0001_r_000000_0
is allowed to commit now
2025-04-14 18:00:57,213 INFO output.FileOutputCommitter: Saved output of task
'attempt_local1682281268_0001_r_000000_0' to hdfs://localhost:54310/output
2025-04-14 18:00:57,214 INFO mapred.LocalJobRunner: Records R/W=49/1 > reduce
2025-04-14 18:00:57,214 INFO mapred.Task: Task
'attempt_local1682281268_0001_r_000000_0' done.
2025-04-14 18:00:57,215 INFO mapred.Task: Final Counters for
attempt_local1682281268_0001_r_000000_0: Counters: 30
	File System Counters
	FILE: Number of bytes read=142517
	FILE: Number of bytes written=787811
	FILE: Number of read operations=0
	FILE: Number of large read operations=0
	FILE: Number of write operations=0
	HDFS: Number of bytes read=338
	HDFS: Number of bytes written=375
	HDFS: Number of read operations=10
	HDFS: Number of large read operations=0
	HDFS: Number of write operations=3
	HDFS: Number of bytes read erasure-coded=0
	Map-Reduce Framework
	Combine input records=0
	Combine output records=0
	Reduce input groups=40
	Reduce shuffle bytes=540
	Reduce input records=49
	Reduce output records=40
```

```
	Spilled Records=49
	Shuffled Maps =1
	Failed Shuffles=0
	Merged Map outputs=1
	GC time elapsed (ms)=0
	Total committed heap usage (bytes)=168820736
	Shuffle Errors
	BAD_ID=0
	CONNECTION=0
	IO_ERROR=0
	WRONG_LENGTH=0
	WRONG_MAP=0
	WRONG_REDUCE=0
	File Output Format Counters
	Bytes Written=375
2025-04-14 18:00:57,216 INFO mapred.LocalJobRunner: Finishing task: attempt_local1682281268_0001_r_000000_0
2025-04-14 18:00:57,216 INFO mapred.LocalJobRunner: reduce task executor complete.
2025-04-14 18:00:57,655 INFO mapreduce.Job:  map 100% reduce 100%
2025-04-14 18:00:57,656 INFO mapreduce.Job: Job job_local1682281268_0001 completed successfully
2025-04-14 18:00:57,690 INFO mapreduce.Job: Counters: 36
	File System Counters
	FILE: Number of bytes read=283922
	FILE: Number of bytes written=1575082
	FILE: Number of read operations=0
	FILE: Number of large read operations=0
	FILE: Number of write operations=0
	HDFS: Number of bytes read=676
	HDFS: Number of bytes written=375
	HDFS: Number of read operations=15
	HDFS: Number of large read operations=0
	HDFS: Number of write operations=4
	HDFS: Number of bytes read erasure-coded=0
	Map-Reduce Framework
	Map input records=1
	Map output records=49
	Map output bytes=436
	Map output materialized bytes=540
	Input split bytes=110
	Combine input records=0
	Combine output records=0
	Reduce input groups=40
	Reduce shuffle bytes=540
	Reduce input records=49
	Reduce output records=40
	Spilled Records=98
	Shuffled Maps =1
	Failed Shuffles=0
	Merged Map outputs=1
	GC time elapsed (ms)=0
	Total committed heap usage (bytes)=337641472
	Shuffle Errors
```

```
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
        File Input Format Counters
        Bytes Read=338
        File Output Format Counters
        Bytes Written=375
2025-04-14 18:00:57,693 INFO streaming.StreamJob: Output
directory: /output
```

9. View Output

```
hduser@Ubuntu:~$ hadoop fs -ls /output
2025-04-14 18:09:15,573 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 hduser supergroup          0 2025-04-14 18:00 /output/_SUCCESS
-rw-r--r--   1 hduser supergroup        375 2025-04-14 18:00 /output/part-00000

hduser@Ubuntu:~$ hadoop fs -cat /output/part-00000
2025-04-14 18:09:34,069 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
(HDFS)    1
Designed  1
Hadoop    2
It   1
MapReduce 1
a    1
across    2
an   1
analysis  1
and  4
applications    1
big  1
clusters  1
computers.      1
data 2
datasets  1
distributed     1
efficient 1
enables   1
fault     1
file 1
for  2
framework 1
in   1
industries.     1
is   2
large     1
model.    1
of   1
```

```
open-source	1
processing	1
programming	1
scalability	1
storing	1
system	1
the	1
tolerance,	1
used	2
uses	1
widely	1
```

```
hduser@Ubuntu:~$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [Ubuntu]
2025-04-14 18:12:00,461 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

```
hduser@Ubuntu:~$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
```

CONCLUSION:-

This practical helps students understand the MapReduce programming model by implementing a simple yet powerful word frequency counter. It demonstrates the use of Hadoop Streaming with Python, and how big data problems can be broken into parallel tasks efficiently.

ASSIGNMENT:- 2

AIM:- Matrix Multiplication

PROBLEM STATEMENT:- Implement Matrix Multiplication using Map-Reduce

THOERY:-

Matrix multiplication is a fundamental operation in linear algebra, commonly used in scientific computing, data analysis, and machine learning. When dealing with large matrices, traditional approaches may not be efficient. To overcome scalability and performance issues, we can leverage Hadoop MapReduce, which allows distributed computation over large datasets.

What is Matrix Multiplication?

Matrix multiplication is an operation where two matrices are multiplied to produce a third matrix. It is defined only when the number of columns in the first matrix A is equal to the number of rows in the second matrix B.

Let:

- Matrix A be of size $m \times n$
- Matrix B be of size $n \times p$

Then, the resulting matrix $C = A \times B$ will be of size $m \times p$.

How Does It Work?

Each element in the resulting matrix $C[i][j]$ is calculated by taking the dot product of the *i-th row* of matrix A and the *j-th column* of matrix B.

$$C_{i,j} = \sum_{k=0}^{n-1} A_{i,k} \times B_{k,j}$$

MapReduce for Matrix Multiplication

1. Mapper Function

The Mapper reads each line of the input matrix and emits intermediate key-value pairs based on the rules of matrix multiplication.

- If the line is from matrix A, the mapper emits intermediate keys for every column in matrix B.
- If the line is from matrix B, the mapper emits intermediate keys for every row in matrix A.

- Key: (i, j) for the resulting matrix
- Value: A tuple indicating source matrix, column/row index, and value

2. Reducer Function

The Reducer receives all values for a given key (i,j). It:

- Separates values from matrix A and B,

- Computes the dot product between the row of A and column of B by multiplying and summing up matching values.

- Output: (i, j, result) - which is the value at position (i, j) of the product matrix.

MapReduce Flow

- Input file is uploaded to HDFS.
- Mapper processes the file and emits intermediate key-value pairs.
- Hadoop shuffles and sorts intermediate data by key.
- Reducer takes each key, computes the final result, and outputs the product matrix elements.

CODE:-

Start Hadoop Services

1. Open terminal and switch to the Hadoop user

```
~$ su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd
```

2. Start HDFS

```
hduser@Ubuntu:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu]
2025-04-17 16:38:54,137 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

3. start-yarn.sh

```
hduser@Ubuntu:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers

hduser@Ubuntu:~$ jps
5393 Jps
4579 DataNode
4820 SecondaryNameNode
4375 NameNode

hduser@Ubuntu:~$ ls
Desktop                     mapper_word_count.py       snap
Documents                   Music                      student_marks.csv
Downloads                   Pictures                   Templates
hadoop-3.3.4                Public                     Videos
hadoop-3.3.4.tar.gz         reducer_student_grade.py  word_count.txt
mapper_student_grade.py  reducer_word_count.py
```

4. Create a text file with 2 matrices

```
hduser@Ubuntu:~$ nano matrix.txt
```

{

matrix.txt file will open in nano input file

}

hduser@Ubuntu:~$ cat matrix.txt

```
A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
B,0,0,5
B,0,1,6
B,1,0,7
B,1,1,8
```

hduser@Ubuntu:~$ hadoop fs -ls /

```
2025-04-17 16:47:01,709 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 4 items
drwxr-xr-x   - hduser supergroup          0 2025-04-15 07:02 /mapreduce_student_grade
drwxr-xr-x   - hduser supergroup          0 2025-04-14 17:34 /mapreduce_word_count
drwxr-xr-x   - hduser supergroup          0 2025-04-14 18:00 /output
drwxr-xr-x   - hduser supergroup          0 2025-04-15 07:32 /output_student_grade
```

5. Create a directory in HDFS and Upload your file

**hduser@Ubuntu:~$ hadoop fs -mkdir /
mapreduce_matrix_multiplication**

```
2025-04-17 16:47:30,213 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /

```
2025-04-17 16:47:39,252 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 5 items
drwxr-xr-x   - hduser supergroup          0 2025-04-17 16:47 /
mapreduce_matrix_multiplication
drwxr-xr-x   - hduser supergroup          0 2025-04-15 07:02 /mapreduce_student_grade
drwxr-xr-x   - hduser supergroup          0 2025-04-14 17:34 /mapreduce_word_count
drwxr-xr-x   - hduser supergroup          0 2025-04-14 18:00 /output
drwxr-xr-x   - hduser supergroup          0 2025-04-15 07:32 /output_student_grade
```

**hduser@Ubuntu:~$ hadoop fs -put matrix.txt /
mapreduce_matrix_multiplication**

```
2025-04-17 16:48:25,553 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /mapreduce_matrix_multiplication

```
2025-04-17 16:48:50,554 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 hduser supergroup         64 2025-04-17 16:48 /
mapreduce_matrix_multiplication/matrix.txt
```

**hduser@Ubuntu:~$ hadoop fs -cat /
mapreduce_matrix_multiplication/matrix.txt**

```
2025-04-17 16:49:24,121 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
A,0,0,1
A,0,1,2
A,1,0,3
A,1,1,4
B,0,0,5
B,0,1,6
B,1,0,7
B,1,1,8
```

6. Create Mapper and Reducer Python Scripts

hduser@Ubuntu:~$ nano mapper_matrix.py

{

After this command mapper_matrix.py will open in nano
Write the mapper code for mapper_matrix

}

hduser@Ubuntu:~$ nano reducer_matrix.py

{

After this command reducer_matrix.py will open in nano
Write the mapper code for reducer_matrix

}

hduser@Ubuntu:~$ chmod +x mapper_matrix.py

hduser@Ubuntu:~$ chmod +x reducer_matrix.py

7. Locate Hadoop Streaming JAR

Now open new terminal (keep the old terminal active) and note the path

~$ su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd

hduser@Ubuntu:~$ cd /usr/local/hadoop/share/hadoop/tools/lib

```
hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ ls
aliyun-java-sdk-core-4.5.10.jar        hadoop-gridmix-3.3.4.jar
aliyun-java-sdk-kms-2.11.0.jar         hadoop-kafka-3.3.4.jar
aliyun-java-sdk-ram-3.1.0.jar          hadoop-minicluster-3.3.4.jar
aliyun-sdk-oss-3.13.0.jar              hadoop-openstack-3.3.4.jar
aws-java-sdk-bundle-1.12.262.jar       hadoop-resourceestimator-3.3.4.jar
azure-data-lake-store-sdk-2.3.9.jar    hadoop-rumen-3.3.4.jar
azure-keyvault-core-1.0.0.jar          hadoop-sls-3.3.4.jar
azure-storage-7.0.1.jar                hadoop-streaming-3.3.4.jar
hadoop-aliyun-3.3.4.jar                hamcrest-core-1.3.jar
hadoop-archive-logs-3.3.4.jar          ini4j-0.5.4.jar
hadoop-archives-3.3.4.jar              jdk.tools-1.8.jar
hadoop-aws-3.3.4.jar                   jdom2-2.0.6.jar
hadoop-azure-3.3.4.jar                 junit-4.13.2.jar
hadoop-azure-datalake-3.3.4.jar        kafka-clients-2.8.1.jar
hadoop-client-3.3.4.jar                lz4-java-1.7.1.jar
hadoop-datajoin-3.3.4.jar              ojalgo-43.0.jar
hadoop-distcp-3.3.4.jar                opentracing-api-0.33.0.jar
hadoop-dynamometer-blockgen-3.3.4.jar opentracing-noop-0.33.0.jar
hadoop-dynamometer-infra-3.3.4.jar     opentracing-util-0.33.0.jar
hadoop-dynamometer-workload-3.3.4.jar org.jacoco.agent-0.8.5-runtime.jar
hadoop-extras-3.3.4.jar                wildfly-openssl-1.0.7.Final.jar
hadoop-fs2img-3.3.4.jar                zstd-jni-1.4.9-1.jar
```

hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ pwd

/usr/local/hadoop/share/hadoop/tools/lib

8. Go back to your old terminal!

Paste the directory got from new terminal

/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar

hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -input /mapreduce_matrix_multiplication/matrix.txt -output /output_matrix -mapper /home/hduser/mapper_matrix.py -reducer /home/hduser/reducer_matrix.py

****Make sure you correctly give your own saved txt and py file names.**

```
2025-04-17 17:04:09,289 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2025-04-17 17:04:10,415 INFO impl.MetricsConfig: Loaded properties from hadoop-
metrics2.properties
2025-04-17 17:04:10,764 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period
at 10 second(s).
2025-04-17 17:04:10,765 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-17 17:04:10,814 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-17 17:04:11,221 INFO mapred.FileInputFormat: Total input files to process : 1
2025-04-17 17:04:11,384 INFO mapreduce.JobSubmitter: number of splits:1
2025-04-17 17:04:11,676 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_local1421426574_0001
2025-04-17 17:04:11,676 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-17 17:04:12,147 INFO mapreduce.Job: The url to track the job: http://
localhost:8080/
2025-04-17 17:04:12,159 INFO mapreduce.Job: Running job: job_local1421426574_0001
2025-04-17 17:04:12,188 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-04-17 17:04:12,191 INFO mapred.LocalJobRunner: OutputCommitter is
org.apache.hadoop.mapred.FileOutputCommitter
2025-04-17 17:04:12,223 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:12,223 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:12,427 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-04-17 17:04:12,437 INFO mapred.LocalJobRunner: Starting task:
attempt_local1421426574_0001_m_000000_0
2025-04-17 17:04:12,518 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:12,518 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:12,597 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-04-17 17:04:12,622 INFO mapred.MapTask: Processing split: hdfs://localhost:54310/
mapreduce_matrix_multiplication/matrix.txt:0+64
2025-04-17 17:04:12,677 INFO mapred.MapTask: numReduceTasks: 1
2025-04-17 17:04:12,812 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-04-17 17:04:12,812 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-04-17 17:04:12,812 INFO mapred.MapTask: soft limit at 83886080
2025-04-17 17:04:12,812 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-04-17 17:04:12,812 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-04-17 17:04:12,816 INFO mapred.MapTask: Map output collector class =
org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-04-17 17:04:12,818 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
mapper_matrix.py]
2025-04-17 17:04:12,826 INFO Configuration.deprecation: mapred.work.output.dir is
deprecated. Instead, use mapreduce.task.output.dir
2025-04-17 17:04:12,827 INFO Configuration.deprecation: mapred.local.dir is
deprecated. Instead, use mapreduce.cluster.local.dir
2025-04-17 17:04:12,828 INFO Configuration.deprecation: map.input.file is deprecated.
Instead, use mapreduce.map.input.file
2025-04-17 17:04:12,828 INFO Configuration.deprecation: map.input.length is
deprecated. Instead, use mapreduce.map.input.length
2025-04-17 17:04:12,828 INFO Configuration.deprecation: mapred.job.id is deprecated.
Instead, use mapreduce.job.id
```

2025-04-17 17:04:12,829 INFO Configuration.deprecation: mapred.task.partition is deprecated. Instead, use mapreduce.task.partition
2025-04-17 17:04:12,832 INFO Configuration.deprecation: map.input.start is deprecated. Instead, use mapreduce.map.input.start
2025-04-17 17:04:12,834 INFO Configuration.deprecation: mapred.task.is.map is deprecated. Instead, use mapreduce.task.ismap
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.tip.id is deprecated. Instead, use mapreduce.task.id
2025-04-17 17:04:12,835 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2025-04-17 17:04:12,841 INFO Configuration.deprecation: user.name is deprecated. Instead, use mapreduce.job.user.name
2025-04-17 17:04:13,050 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec/s]
2025-04-17 17:04:13,058 INFO streaming.PipeMapRed: Records R/W=8/1
2025-04-17 17:04:13,061 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-17 17:04:13,062 INFO streaming.PipeMapRed: mapRedFinished
2025-04-17 17:04:13,072 INFO mapred.LocalJobRunner:
2025-04-17 17:04:13,075 INFO mapred.MapTask: Starting flush of map output
2025-04-17 17:04:13,076 INFO mapred.MapTask: Spilling map output
2025-04-17 17:04:13,076 INFO mapred.MapTask: bufstart = 0; bufend = 192; bufvoid = 104857600
2025-04-17 17:04:13,076 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214336(104857344); length = 61/6553600
2025-04-17 17:04:13,117 INFO mapred.MapTask: Finished spill 0
2025-04-17 17:04:13,149 INFO mapred.Task: Task:attempt_local1421426574_0001_m_000000_0 is done. And is in the process of committing
2025-04-17 17:04:13,158 INFO mapred.LocalJobRunner: Records R/W=8/1
2025-04-17 17:04:13,160 INFO mapred.Task: Task 'attempt_local1421426574_0001_m_000000_0' done.
2025-04-17 17:04:13,182 INFO mapred.Task: Final Counters for attempt_local1421426574_0001_m_000000_0: **Counters: 23**

File System Counters
FILE: Number of bytes read=141410
FILE: Number of bytes written=786978
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=64
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=8
Map output records=16
Map output bytes=192
Map output materialized bytes=230
Input split bytes=117
Combine input records=0
Spilled Records=16
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=0
Total committed heap usage (bytes)=168820736
File Input Format Counters
Bytes Read=64

2025-04-17 17:04:13,183 INFO mapred.LocalJobRunner: Finishing task: attempt_local1421426574_0001_m_000000_0

```
2025-04-17 17:04:13,184 INFO mapred.LocalJobRunner: map task executor complete.
2025-04-17 17:04:13,187 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2025-04-17 17:04:13,187 INFO mapred.LocalJobRunner: Starting task:
attempt_local1421426574_0001_r_000000_0
2025-04-17 17:04:13,196 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-17 17:04:13,196 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-17 17:04:13,196 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-04-17 17:04:13,202 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin:
org.apache.hadoop.mapreduce.task.reduce.Shuffle@310aa1b6
2025-04-17 17:04:13,206 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-17 17:04:13,216 INFO mapreduce.Job: Job job_local1421426574_0001 running in
uber mode : false
2025-04-17 17:04:13,218 INFO mapreduce.Job:  map 100% reduce 0%
2025-04-17 17:04:13,247 INFO reduce.MergeManagerImpl: MergerManager:
memoryLimit=546098368, maxSingleShuffleLimit=136524592, mergeThreshold=360424928,
ioSortFactor=10, memToMemMergeOutputsThreshold=10
2025-04-17 17:04:13,252 INFO reduce.EventFetcher:
attempt_local1421426574_0001_r_000000_0 Thread started: EventFetcher for fetching Map
Completion Events
2025-04-17 17:04:13,310 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle
output of map attempt_local1421426574_0001_m_000000_0 decomp: 226 len: 230 to MEMORY
2025-04-17 17:04:13,318 INFO reduce.InMemoryMapOutput: Read 226 bytes from map-output
for attempt_local1421426574_0001_m_000000_0
2025-04-17 17:04:13,327 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output
of size: 226, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->226
2025-04-17 17:04:13,333 INFO reduce.EventFetcher: EventFetcher is interrupted..
Returning
2025-04-17 17:04:13,338 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,340 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-
memory map-outputs and 0 on-disk map-outputs
2025-04-17 17:04:13,368 INFO mapred.Merger: Merging 1 sorted segments
2025-04-17 17:04:13,368 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 220 bytes
2025-04-17 17:04:13,379 INFO reduce.MergeManagerImpl: Merged 1 segments, 226 bytes to
disk to satisfy reduce memory limit
2025-04-17 17:04:13,381 INFO reduce.MergeManagerImpl: Merging 1 files, 230 bytes from
disk
2025-04-17 17:04:13,382 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from
memory into reduce
2025-04-17 17:04:13,382 INFO mapred.Merger: Merging 1 sorted segments
2025-04-17 17:04:13,382 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 220 bytes
2025-04-17 17:04:13,383 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,386 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
reducer_matrix.py]
2025-04-17 17:04:13,397 INFO Configuration.deprecation: mapred.job.tracker is
deprecated. Instead, use mapreduce.jobtracker.address
2025-04-17 17:04:13,399 INFO Configuration.deprecation: mapred.map.tasks is
deprecated. Instead, use mapreduce.job.maps
2025-04-17 17:04:13,505 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-17 17:04:13,508 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-17 17:04:13,513 INFO streaming.PipeMapRed: Records R/W=16/1
2025-04-17 17:04:13,516 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-17 17:04:13,517 INFO streaming.PipeMapRed: mapRedFinished
2025-04-17 17:04:13,629 INFO mapred.Task: Task:attempt_local1421426574_0001_r_000000_0
is done. And is in the process of committing
2025-04-17 17:04:13,637 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-17 17:04:13,637 INFO mapred.Task: Task attempt_local1421426574_0001_r_000000_0
is allowed to commit now
2025-04-17 17:04:13,705 INFO output.FileOutputCommitter: Saved output of task
'attempt_local1421426574_0001_r_000000_0' to hdfs://localhost:54310/output_matrix
2025-04-17 17:04:13,709 INFO mapred.LocalJobRunner: Records R/W=16/1 > reduce
2025-04-17 17:04:13,709 INFO mapred.Task: Task
'attempt_local1421426574_0001_r_000000_0' done.
2025-04-17 17:04:13,710 INFO mapred.Task: Final Counters for
attempt_local1421426574_0001_r_000000_0: Counters: 30
```

```
	File System Counters
	FILE: Number of bytes read=141902
	FILE: Number of bytes written=787208
	FILE: Number of read operations=0
	FILE: Number of large read operations=0
	FILE: Number of write operations=0
	HDFS: Number of bytes read=64
	HDFS: Number of bytes written=36
	HDFS: Number of read operations=10
	HDFS: Number of large read operations=0
	HDFS: Number of write operations=3
	HDFS: Number of bytes read erasure-coded=0
	Map-Reduce Framework
	Combine input records=0
	Combine output records=0
	Reduce input groups=4
	Reduce shuffle bytes=230
	Reduce input records=16
	Reduce output records=4
	Spilled Records=16
	Shuffled Maps =1
	Failed Shuffles=0
	Merged Map outputs=1
	GC time elapsed (ms)=0
	Total committed heap usage (bytes)=168820736
	Shuffle Errors
	BAD_ID=0
	CONNECTION=0
	IO_ERROR=0
	WRONG_LENGTH=0
	WRONG_MAP=0
	WRONG_REDUCE=0
	File Output Format Counters
	Bytes Written=36
2025-04-17 17:04:13,710 INFO mapred.LocalJobRunner: Finishing task:
attempt_local1421426574_0001_r_000000_0
2025-04-17 17:04:13,710 INFO mapred.LocalJobRunner: reduce task executor complete.
2025-04-17 17:04:14,222 INFO mapreduce.Job:  map 100% reduce 100%
2025-04-17 17:04:14,223 INFO mapreduce.Job: Job job_local1421426574_0001 completed
successfully
2025-04-17 17:04:14,253 INFO mapreduce.Job: Counters: 36
	File System Counters
	FILE: Number of bytes read=283312
	FILE: Number of bytes written=1574186
	FILE: Number of read operations=0
	FILE: Number of large read operations=0
	FILE: Number of write operations=0
	HDFS: Number of bytes read=128
	HDFS: Number of bytes written=36
	HDFS: Number of read operations=15
	HDFS: Number of large read operations=0
	HDFS: Number of write operations=4
	HDFS: Number of bytes read erasure-coded=0
```

```
	Map-Reduce Framework
	Map input records=8
	Map output records=16
	Map output bytes=192
	Map output materialized bytes=230
	Input split bytes=117
	Combine input records=0
	Combine output records=0
	Reduce input groups=4
	Reduce shuffle bytes=230
	Reduce input records=16
	Reduce output records=4
	Spilled Records=32
	Shuffled Maps =1
	Failed Shuffles=0
	Merged Map outputs=1
	GC time elapsed (ms)=0
	Total committed heap usage (bytes)=337641472
	Shuffle Errors
	BAD_ID=0
	CONNECTION=0
	IO_ERROR=0
	WRONG_LENGTH=0
	WRONG_MAP=0
	WRONG_REDUCE=0
	File Input Format Counters
	Bytes Read=64
	File Output Format Counters
	Bytes Written=36
2025-04-17 17:04:14,254 INFO streaming.StreamJob: Output
directory: /output_matrix
```

9. View Output

```
hduser@Ubuntu:~$ hadoop fs -ls /output_matrix
2025-04-17 17:11:31,287 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 hduser supergroup          0 2025-04-17 17:04 /output_matrix/
_SUCCESS
-rw-r--r--   1 hduser supergroup         36 2025-04-17 17:04 /output_matrix/
part-00000
```

```
hduser@Ubuntu:~$ hadoop fs -cat /output_matrix/part-00000
2025-04-17 17:11:45,160 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
0,0	19.0
0,1	22.0
1,0	43.0
1,1	50.0
```

CONCLUSION:-

Matrix multiplication using Hadoop MapReduce is a classic example of dividing a computational task across nodes in a distributed system. This approach showcases the scalability and efficiency of Hadoop when dealing with large matrix data, and demonstrates how traditional algorithms can be restructured for distributed frameworks.

ASSIGNMENT:- 3

AIM:- Finding Grades of Students

PROBLEM STATEMENT:- Develop a MapReduce program to find the grades of students.

THOERY:-

What is Hadoop?

Apache Hadoop is an open-source, Java-based software framework that allows for the distributed storage and processing of large datasets using clusters of computers. It was created by Doug Cutting and Mike Cafarella and inspired by Google's MapReduce and Google File System (GFS) papers.

It is designed to scale from a single server to thousands of machines, each offering local computation and storage.

Hadoop Core Modules

Hadoop is made up of the following three main components:

1. Hadoop Distributed File System (HDFS)
2. MapReduce (Processing Model)
3. YARN (Yet Another Resource Negotiator)

- **HDFS (Hadoop Distributed File System)**

HDFS is a distributed file system designed to store very large data sets reliably and stream those data sets at high bandwidth to user applications.

Key Features:

- Master-slave architecture with NameNode and DataNodes.
- Files are split into blocks (default 128 MB or 256 MB).
- Blocks are replicated across DataNodes for fault tolerance (default: 3 replicas).
- Designed for batch processing rather than real-time.

Components:

- **NameNode:** Manages the file system namespace and controls access.
- **DataNode:** Stores actual data blocks.

- **YARN (Yet Another Resource Negotiator)**

YARN is the resource management layer of Hadoop. It allows multiple data processing engines like MapReduce, Spark, Tez, etc., to run and process data stored in HDFS.

Components:

- **ResourceManager (RM):** Manages resources and job scheduling.
- **NodeManager (NM):** Monitors resource usage on each node.
- **ApplicationMaster (AM):** Manages the lifecycle of a single application.

- **MapReduce (Computation Model)**

MapReduce is a programming model used for processing large datasets in parallel across a Hadoop cluster.

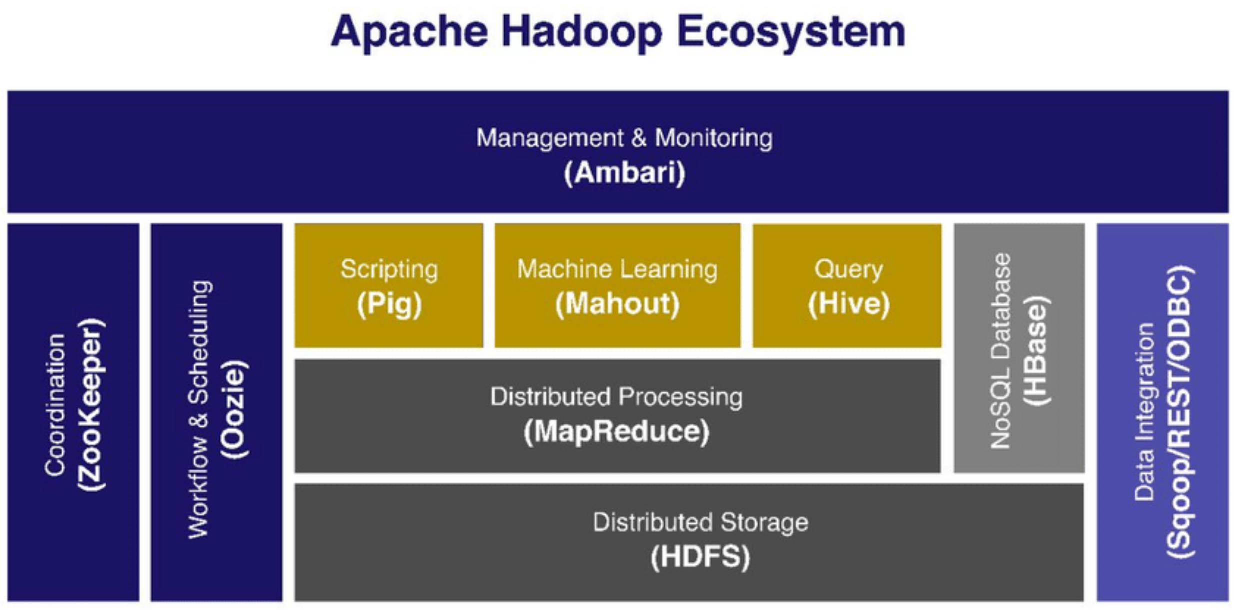
Stages of MapReduce:

A. **Input Splits:** The input is divided into manageable splits for processing.

B. Mapper Phase:
- Each mapper works on one split of data.
- Outputs intermediate key-value pairs.

C. **Shuffle & Sort:**
- All intermediate key-value pairs are shuffled to group values by keys.
- Sorted so that all values for a key are sent to the same reducer.

D. **Reducer Phase:**
- Aggregates values for each key.
- Outputs the final result.



Procedure

1. **Input Format:** Takes raw input data and splits it into lines/records that mappers can process.
2. **Mapper:** Processes each record and outputs intermediate (key, value) pairs. **Example:** Line: "8018,BDA,90" Output: ("8018", 90)
3. **Combiner (Optional):** Acts as a mini-reducer that runs after the mapper to reduce the volume of data transferred to the reducer.
4. **Partitioner:** Decides which reducer will get which key-value pair. Usually based on hashing of keys.
5. **Shuffle and Sort:** Intermediate keys are sorted and grouped. Ensures that each reducer gets all values for a given key.
6. **Reducer:** Accepts key and list of values, applies logic to compute final result.
7. **Output Format:** Writes the final output (e.g., student_id, average, grade) to HDFS.

CODE:-

1. Open terminal and switch to the Hadoop user

```
yash_zagekar@Ubuntu:~$ su hduser

hduser@Ubuntu:/home/yash_zagekar$ cd
```

2. Start HDFS

```
hduser@Ubuntu:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu]
2025-04-15 06:20:16,059 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

```
hduser@Ubuntu:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

```
hduser@Ubuntu:~$ jps
3313 SecondaryNameNode
5001 Jps
3115 DataNode
2973 NameNode
```

3. Creating student_marks.csv

```
hduser@Ubuntu:~$ ls
Desktop       hadoop-3.3.4.tar.gz   Public                    Videos
Documents     mapper_word_count.py  reducer_word_count.py  word_count.txt
Downloads     Music                 snap
hadoop-3.3.4   Pictures             Templates
```

```
hduser@Ubuntu:~$ nano student_marks.csv
```

{

Student_marks.csv will open in nano

Input your data in csv file

}

```
hduser@Ubuntu:~$ cat student_marks.csv
student_id,subject,marks
8018,BI,85
8018,BDA,90
8018,CI,78
8018,DC,93
8028,BI,97
8028,BDA,99
8028,CI,95
8028,DC,90
8032,BI,86
8032,BDA,94
8032,CI,85
8032,DC,96
8034,BI,60
8034,BDA,55
8034,CI,40
8034,DC,40
8095,BI,85
8095,BDA,90
8095,CI,96
8095,DC,97
```

4. Creating directory and putting student_marks.csv inside it

```
hduser@Ubuntu:~$ hadoop fs -mkdir /mapreduce_student_grade
```

```
2025-04-15 07:01:52,065 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /

```
2025-04-15 07:02:02,774 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
drwxr-xr-x   - hduser supergroup          0 2025-04-15 07:01 /mapreduce_student_grade
drwxr-xr-x   - hduser supergroup          0 2025-04-14 17:34 /mapreduce_word_count
drwxr-xr-x   - hduser supergroup          0 2025-04-14 18:00 /output
```

hduser@Ubuntu:~$ hadoop fs -put student_marks.csv /mapreduce_student_grade

```
2025-04-15 07:02:42,787 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

hduser@Ubuntu:~$ hadoop fs -ls /mapreduce_student_grade

```
2025-04-15 07:02:57,300 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 hduser supergroup        250 2025-04-15 07:02 /mapreduce_student_grade/student_marks.csv
```

hduser@Ubuntu:~$ hadoop fs -cat /mapreduce_student_grade/student_marks.csv

```
2025-04-15 07:07:25,530 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
student_id,subject,marks
8018,BI,85
8018,BDA,90
8018,CI,78
8018,DC,93
8028,BI,97
8028,BDA,99
8028,CI,95
8028,DC,90
8032,BI,86
8032,BDA,94
8032,CI,85
8032,DC,96
8034,BI,60
8034,BDA,55
8034,CI,40
8034,DC,40
8095,BI,85
8095,BDA,90
8095,CI,96
8095,DC,97
```

5. Creating mapper_student_grade.py and reducer_student_grade.py

hduser@Ubuntu:~$ nano mapper_student_grade.py

{

This will open mapper_student_grade.py in nano, write mapper code

}

hduser@Ubuntu:~$ nano reducer_student_grade.py

{

This will open reducer_student_grade.py in nano, write reducer code

}

`hduser@Ubuntu:~$ chmod +x mapper_student_grade.py`

`hduser@Ubuntu:~$ chmod +x reducer_student_grade.py`

6. Locate Hadoop Streaming JAR

****Now open new terminal (keep the old terminal active)**

`~$ su hduser`

`hduser@Ubuntu:/home/yash_zagekar$ cd`

`hduser@Ubuntu:~$ cd /usr/local/hadoop/share/hadoop/tools/lib`

`hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ ls`

```
aliyun-java-sdk-core-4.5.10.jar          hadoop-aws-3.3.4.jar
hadoop-gridmix-3.3.4.jar                 jdom2-2.0.6.jar
aliyun-java-sdk-kms-2.11.0.jar           hadoop-azure-3.3.4.jar
hadoop-kafka-3.3.4.jar                   junit-4.13.2.jar
aliyun-java-sdk-ram-3.1.0.jar            hadoop-azure-datalake-3.3.4.jar
hadoop-minicluster-3.3.4.jar             kafka-clients-2.8.1.jar
aliyun-sdk-oss-3.13.0.jar                hadoop-client-3.3.4.jar
hadoop-openstack-3.3.4.jar               lz4-java-1.7.1.jar
aws-java-sdk-bundle-1.12.262.jar         hadoop-datajoin-3.3.4.jar
hadoop-resourceestimator-3.3.4.jar       ojalgo-43.0.jar
azure-data-lake-store-sdk-2.3.9.jar      hadoop-distcp-3.3.4.jar
hadoop-rumen-3.3.4.jar                   opentracing-api-0.33.0.jar
azure-keyvault-core-1.0.0.jar            hadoop-dynamometer-blockgen-3.3.4.jar
hadoop-sls-3.3.4.jar                     opentracing-noop-0.33.0.jar
azure-storage-7.0.1.jar                  hadoop-dynamometer-infra-3.3.4.jar
hadoop-streaming-3.3.4.jar               opentracing-util-0.33.0.jar
hadoop-aliyun-3.3.4.jar                  hadoop-dynamometer-workload-3.3.4.jar
hamcrest-core-1.3.jar                    org.jacoco.agent-0.8.5-runtime.jar
hadoop-archive-logs-3.3.4.jar            hadoop-extras-3.3.4.jar
ini4j-0.5.4.jar                          wildfly-openssl-1.0.7.Final.jar
hadoop-archives-3.3.4.jar                hadoop-fs2img-3.3.4.jar
jdk.tools-1.8.jar                        zstd-jni-1.4.9-1.jar
```

`hduser@Ubuntu:/usr/local/hadoop/share/hadoop/tools/lib$ pwd`

`/usr/local/hadoop/share/hadoop/tools/lib`

7. Go back to your old terminal!

Paste the directory got from new terminal

`/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar`

hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -input /mapreduce_student_grade/student_marks.csv -output /output_student_grade -mapper /home/hduser/mapper_student_grade.py -reducer /home/hduser/reducer_student_grade.py

```
2025-04-15 07:32:09,423 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
2025-04-15 07:32:11,296 INFO impl.MetricsConfig: Loaded properties from hadoop-
metrics2.properties
2025-04-15 07:32:11,836 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period
at 10 second(s).
2025-04-15 07:32:11,836 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-15 07:32:11,901 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-15 07:32:12,632 INFO mapred.FileInputFormat: Total input files to process : 1
2025-04-15 07:32:12,931 INFO mapreduce.JobSubmitter: number of splits:1
2025-04-15 07:32:13,518 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_local1541048102_0001
2025-04-15 07:32:13,518 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-15 07:32:13,943 INFO mapreduce.Job: The url to track the job: http://
localhost:8080/
2025-04-15 07:32:13,996 INFO mapreduce.Job: Running job: job_local1541048102_0001
2025-04-15 07:32:13,998 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-04-15 07:32:14,019 INFO mapred.LocalJobRunner: OutputCommitter is
org.apache.hadoop.mapred.FileOutputCommitter
2025-04-15 07:32:14,051 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-15 07:32:14,052 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-15 07:32:14,285 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-04-15 07:32:14,302 INFO mapred.LocalJobRunner: Starting task:
attempt_local1541048102_0001_m_000000_0
2025-04-15 07:32:14,417 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-15 07:32:14,417 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-15 07:32:14,520 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-04-15 07:32:14,564 INFO mapred.MapTask: Processing split: hdfs://localhost:54310/
mapreduce_student_grade/student_marks.csv:0+250
2025-04-15 07:32:14,655 INFO mapred.MapTask: numReduceTasks: 1
2025-04-15 07:32:14,880 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-04-15 07:32:14,881 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-04-15 07:32:14,888 INFO mapred.MapTask: soft limit at 83886080
2025-04-15 07:32:14,889 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-04-15 07:32:14,889 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-04-15 07:32:14,908 INFO mapred.MapTask: Map output collector class =
org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-04-15 07:32:14,923 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
mapper_student_grade.py]
2025-04-15 07:32:14,945 INFO Configuration.deprecation: mapred.work.output.dir is
deprecated. Instead, use mapreduce.task.output.dir
2025-04-15 07:32:14,957 INFO Configuration.deprecation: mapred.local.dir is
deprecated. Instead, use mapreduce.cluster.local.dir
2025-04-15 07:32:14,961 INFO Configuration.deprecation: map.input.file is deprecated.
Instead, use mapreduce.map.input.file
```

```
2025-04-15 07:32:14,962 INFO Configuration.deprecation: map.input.length is deprecated. Instead, use mapreduce.map.input.length
2025-04-15 07:32:14,962 INFO Configuration.deprecation: mapred.job.id is deprecated. Instead, use mapreduce.job.id
2025-04-15 07:32:14,967 INFO Configuration.deprecation: mapred.task.partition is deprecated. Instead, use mapreduce.task.partition
2025-04-15 07:32:14,969 INFO Configuration.deprecation: map.input.start is deprecated. Instead, use mapreduce.map.input.start
2025-04-15 07:32:14,969 INFO Configuration.deprecation: mapred.task.is.map is deprecated. Instead, use mapreduce.task.ismap
2025-04-15 07:32:14,970 INFO Configuration.deprecation: mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2025-04-15 07:32:14,970 INFO Configuration.deprecation: mapred.tip.id is deprecated. Instead, use mapreduce.task.id
2025-04-15 07:32:14,970 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2025-04-15 07:32:14,971 INFO Configuration.deprecation: user.name is deprecated. Instead, use mapreduce.job.user.name
2025-04-15 07:32:15,072 INFO mapreduce.Job: Job job_local1541048102_0001 running in uber mode : false
2025-04-15 07:32:15,081 INFO mapreduce.Job:  map 0% reduce 0%
2025-04-15 07:32:15,461 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA [rec/s]
2025-04-15 07:32:15,461 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA [rec/s]
2025-04-15 07:32:15,474 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-15 07:32:15,478 INFO streaming.PipeMapRed: Records R/W=21/1
2025-04-15 07:32:15,480 INFO streaming.PipeMapRed: mapRedFinished
2025-04-15 07:32:15,529 INFO mapred.LocalJobRunner:
2025-04-15 07:32:15,529 INFO mapred.MapTask: Starting flush of map output
2025-04-15 07:32:15,529 INFO mapred.MapTask: Spilling map output
2025-04-15 07:32:15,529 INFO mapred.MapTask: bufstart = 0; bufend = 160; bufvoid = 104857600
2025-04-15 07:32:15,529 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214320(104857280); length = 77/6553600
2025-04-15 07:32:15,613 INFO mapred.MapTask: Finished spill 0
2025-04-15 07:32:15,689 INFO mapred.Task: Task:attempt_local1541048102_0001_m_000000_0 is done. And is in the process of committing
2025-04-15 07:32:15,739 INFO mapred.LocalJobRunner: Records R/W=21/1
2025-04-15 07:32:15,740 INFO mapred.Task: Task 'attempt_local1541048102_0001_m_000000_0' done.
2025-04-15 07:32:15,785 INFO mapred.Task: Final Counters for attempt_local1541048102_0001_m_000000_0: Counters: 23
```

File System Counters

FILE: Number of bytes read=141410

FILE: Number of bytes written=786994

FILE: Number of read operations=0

FILE: Number of large read operations=0

FILE: Number of write operations=0

HDFS: Number of bytes read=250

HDFS: Number of bytes written=0

HDFS: Number of read operations=5

HDFS: Number of large read operations=0

HDFS: Number of write operations=1

HDFS: Number of bytes read erasure-coded=0

Map-Reduce Framework

Map input records=21

```
        Map output records=20
        Map output bytes=160
        Map output materialized bytes=206
        Input split bytes=116
        Combine input records=0
        Spilled Records=20
        Failed Shuffles=0
        Merged Map outputs=0
        GC time elapsed (ms)=0
        Total committed heap usage (bytes)=176160768
        File Input Format Counters
        Bytes Read=250
2025-04-15 07:32:15,799 INFO mapred.LocalJobRunner: Finishing task:
attempt_local1541048102_0001_m_000000_0
2025-04-15 07:32:15,810 INFO mapred.LocalJobRunner: map task executor complete.
2025-04-15 07:32:15,824 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2025-04-15 07:32:15,825 INFO mapred.LocalJobRunner: Starting task:
attempt_local1541048102_0001_r_000000_0
2025-04-15 07:32:15,881 INFO output.FileOutputCommitter: File Output Committer
Algorithm version is 2
2025-04-15 07:32:15,881 INFO output.FileOutputCommitter: FileOutputCommitter skip
cleanup _temporary folders under output directory:false, ignore cleanup failures:
false
2025-04-15 07:32:15,885 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-04-15 07:32:15,906 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin:
org.apache.hadoop.mapreduce.task.reduce.Shuffle@16ba503b
2025-04-15 07:32:15,913 WARN impl.MetricsSystemImpl: JobTracker metrics system already
initialized!
2025-04-15 07:32:15,997 INFO reduce.MergeManagerImpl: MergerManager:
memoryLimit=546098368, maxSingleShuffleLimit=136524592, mergeThreshold=360424928,
ioSortFactor=10, memToMemMergeOutputsThreshold=10
2025-04-15 07:32:16,003 INFO reduce.EventFetcher:
attempt_local1541048102_0001_r_000000_0 Thread started: EventFetcher for fetching Map
Completion Events
2025-04-15 07:32:16,095 INFO mapreduce.Job:  map 100% reduce 0%
2025-04-15 07:32:16,176 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle
output of map attempt_local1541048102_0001_m_000000_0 decomp: 202 len: 206 to MEMORY
2025-04-15 07:32:16,214 INFO reduce.InMemoryMapOutput: Read 202 bytes from map-output
for attempt_local1541048102_0001_m_000000_0
2025-04-15 07:32:16,231 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output
of size: 202, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->202
2025-04-15 07:32:16,256 INFO reduce.EventFetcher: EventFetcher is interrupted..
Returning
2025-04-15 07:32:16,263 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-15 07:32:16,264 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-
memory map-outputs and 0 on-disk map-outputs
2025-04-15 07:32:16,323 INFO mapred.Merger: Merging 1 sorted segments
2025-04-15 07:32:16,324 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 195 bytes
2025-04-15 07:32:16,343 INFO reduce.MergeManagerImpl: Merged 1 segments, 202 bytes to
disk to satisfy reduce memory limit
2025-04-15 07:32:16,347 INFO reduce.MergeManagerImpl: Merging 1 files, 206 bytes from
disk
2025-04-15 07:32:16,349 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from
memory into reduce
2025-04-15 07:32:16,351 INFO mapred.Merger: Merging 1 sorted segments
2025-04-15 07:32:16,352 INFO mapred.Merger: Down to the last merge-pass, with 1
segments left of total size: 195 bytes
```

```
2025-04-15 07:32:16,355 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-15 07:32:16,366 INFO streaming.PipeMapRed: PipeMapRed exec [/home/hduser/
reducer_student_grade.py]
2025-04-15 07:32:16,377 INFO Configuration.deprecation: mapred.job.tracker is
deprecated. Instead, use mapreduce.jobtracker.address
2025-04-15 07:32:16,382 INFO Configuration.deprecation: mapred.map.tasks is
deprecated. Instead, use mapreduce.job.maps
2025-04-15 07:32:16,688 INFO streaming.PipeMapRed: R/W/S=1/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-15 07:32:16,689 INFO streaming.PipeMapRed: R/W/S=10/0/0 in:NA [rec/s] out:NA
[rec/s]
2025-04-15 07:32:16,694 INFO streaming.PipeMapRed: Records R/W=20/1
2025-04-15 07:32:16,704 INFO streaming.PipeMapRed: MRErrorThread done
2025-04-15 07:32:16,705 INFO streaming.PipeMapRed: mapRedFinished
2025-04-15 07:32:16,953 INFO mapred.Task: Task:attempt_local1541048102_0001_r_000000_0
is done. And is in the process of committing
2025-04-15 07:32:16,966 INFO mapred.LocalJobRunner: 1 / 1 copied.
2025-04-15 07:32:16,967 INFO mapred.Task: Task attempt_local1541048102_0001_r_000000_0
is allowed to commit now
2025-04-15 07:32:17,108 INFO output.FileOutputCommitter: Saved output of task
'attempt_local1541048102_0001_r_000000_0' to hdfs://localhost:54310/
output_student_grade
2025-04-15 07:32:17,112 INFO mapred.LocalJobRunner: Records R/W=20/1 > reduce
2025-04-15 07:32:17,114 INFO mapred.Task: Task
'attempt_local1541048102_0001_r_000000_0' done.
2025-04-15 07:32:17,115 INFO mapred.Task: Final Counters for
attempt_local1541048102_0001_r_000000_0: Counters: 30
	File System Counters
	FILE: Number of bytes read=141854
	FILE: Number of bytes written=787200
	FILE: Number of read operations=0
	FILE: Number of large read operations=0
	FILE: Number of write operations=0
	HDFS: Number of bytes read=250
	HDFS: Number of bytes written=65
	HDFS: Number of read operations=10
	HDFS: Number of large read operations=0
	HDFS: Number of write operations=3
	HDFS: Number of bytes read erasure-coded=0
	Map-Reduce Framework
	Combine input records=0
	Combine output records=0
	Reduce input groups=5
	Reduce shuffle bytes=206
	Reduce input records=20
	Reduce output records=5
	Spilled Records=20
	Shuffled Maps =1
	Failed Shuffles=0
	Merged Map outputs=1
	GC time elapsed (ms)=127
	Total committed heap usage (bytes)=181403648
```

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=65

```
2025-04-15 07:32:17,117 INFO mapred.LocalJobRunner: Finishing task:
attempt_local1541048102_0001_r_000000_0
2025-04-15 07:32:17,118 INFO mapred.LocalJobRunner: reduce task executor complete.
2025-04-15 07:32:18,104 INFO mapreduce.Job:  map 100% reduce 100%
2025-04-15 07:32:18,105 INFO mapreduce.Job: Job job_local1541048102_0001 completed
successfully
2025-04-15 07:32:18,194 INFO mapreduce.Job: Counters: 36
```

File System Counters
FILE: Number of bytes read=283264
FILE: Number of bytes written=1574194
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=500
HDFS: Number of bytes written=65
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=21
Map output records=20
Map output bytes=160
Map output materialized bytes=206
Input split bytes=116
Combine input records=0
Combine output records=0
Reduce input groups=5
Reduce shuffle bytes=206
Reduce input records=20
Reduce output records=5
Spilled Records=40
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=127
Total committed heap usage (bytes)=357564416
Shuffle Errors

```
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=250
File Output Format Counters
Bytes Written=65
2025-04-15 07:32:18,197 INFO streaming.StreamJob: Output directory: /output_student_grade
```

9. View Output

```
hduser@Ubuntu:~$ hadoop fs -ls /output_student_grade
2025-04-15 07:32:53,798 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 hduser supergroup          0 2025-04-15 07:32 /output_student_grade/_SUCCESS
-rw-r--r--   1 hduser supergroup         65 2025-04-15 07:32 /output_student_grade/part-00000

hduser@Ubuntu:~$ hadoop fs -cat /output_student_grade/part-00000
2025-04-15 07:33:28,617 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
8018 86.50      B
8028 95.25      A
8032 90.25      A
8034 48.75      F
8095 92.00      A

hduser@Ubuntu:~$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [Ubuntu]
2025-04-14 18:12:00,461 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

hduser@Ubuntu:~$ stop-yarn.sh
```

CONCLUSION:-

This practical demonstrates how Hadoop's MapReduce paradigm can be effectively used to process large-scale student data to derive meaningful insights like average and grade. By distributing the workload between Mappers and Reducers, the computation becomes efficient and scalable.

ASSIGNMENT:- 4

AIM:- MongoDB CRUD Operations

PROBLEM STATEMENT:- Creation of database and CRUD Operations: Insert, Update and Delete Document and Query Operations.

THOERY:-

Introduction to MongoDB

MongoDB is a NoSQL (Non-relational), document-oriented database developed by MongoDB Inc. It is designed for ease of development and scaling, and is used for handling large volumes of unstructured or semi-structured data. Unlike traditional Relational Database Management Systems (RDBMS) that store data in rows and tables, MongoDB stores data in JSON-like documents, which makes it highly flexible.

Documents

- Documents are the basic unit of data in MongoDB. Each document is a set of key-value pairs.
- Documents are similar to JSON (JavaScript Object Notation), but use a binary format called BSON (Binary JSON), which allows for more data types and efficient storage.

Example of a document:

```
{
  "name": "John",
  "age": 21,
  "course": "Data Science",
  "skills": ["Python", "MongoDB", "Machine Learning"]
}
```

MongoDB creates a database implicitly when data is inserted. To switch to a database:

```
use database_name;
```

Collections

- A collection is a group of MongoDB documents. Collections are similar to tables in RDBMS.
- Unlike SQL tables, documents in a collection do not require a fixed schema, meaning each document can have different fields.

A collection is created when a document is inserted or explicitly using:

```
db.createCollection("collection_name");
```

Feature	MongoDB	RDBMS (e.g., MySQL, PostgreSQL)
Data Storage	Stores data as documents in BSON/ JSON-like format. Each document is a self-contained unit.	Stores data in tables with rows and columns.
Schema	Dynamic schema: No need to define the structure in advance. Fields can differ per document.	Fixed schema: Must define the table structure before inserting data.
Joins	Supports limited joins using `$lookup`, but it's not as native or powerful as in SQL.	Fully supports JOIN operations across multiple tables.
Scalability	Horizontally scalable via sharding (data is distributed across multiple servers).	Typically vertically scalable (scale by upgrading hardware).
Transactions	Supports multi-document ACID transactionssince v4.0.	Fully supports ACID-compliant transactions across multiple tables.
Use Case Fit	Great for unstructured, semi-structured, and rapidly changing data.	Best for structured data, strong relational logic, and complex queries.

CRUD Operations in MongoDB

CRUD stands for Create, Read, Update, and Delete — the four fundamental operations performed on any data in a database. MongoDB provides simple and powerful methods to perform these operations on its document-based collections.

1. **Create – Insert Data:-** MongoDB uses the insertOne() and insertMany() methods to add documents into a collection.
   1. **insertOne():-** Used to insert a single document into a collection.
      Syntax:
      ```
      db.collection_name.insertOne({key1: value1, key2: value2, ...});
      ```
   2. **insertMany():-** Used to insert multiple documents at once.
      Syntax:
      ```
      db.collection_name.insertMany([{...}, {...}, ...]);
      ```

2. **Read – Query Data:-** To retrieve data from MongoDB, the find() method is used.
   1. **find():-** Fetches documents from a collection. If no condition is provided, it returns all documents.
      Examples:
      - `db.products.find();  // Fetches all documents`
      - `db.products.find({name: “Notebook"}); // Fetches documents with name = "Notebook"`

3. **Update – Modify Data:-** updateOne() or updateMany() functions help in modify ing one or more documents.
   1. **updateOne():-** Updates the first matching document.
      Syntax:

```
db.collection_name.updateOne(filter, update_operation);
```

2. **updateMany():-** Updates all documents that match the condition.
   Syntax:
   ```
   db.collection_name.updateMany(filter, update_operation);;
   ```

4. **Delete – Remove Data:-** Delete data using deleteOne() or deleteMany().
   1. **deleteOne():-** Removes the first document that matches the condition.
      Example:
      ```
      db.products.deleteOne({name: "Laptop"});
      ```
   2. **deleteMany():-** Deletes all documents that match the condition.
      Examples:
      - `db.products.deleteMany({category: "Books"});  // Delete all books`
      - `db.products.deleteMany({});  // Deletes ALL documents from the collection!`

5. **Query Operators in MongoDB**

MongoDB offers a wide range of comparison and logical operators to build complex queries:

1. Comparison Operators

Operator	Description
`$eq`	Equals
`$ne`	Not equal
`$gt`	Greater than
`$lt`	Less than
`$gte`	Greater than or equal to
`$lte`	Less than or equal to

2. Logical Operators

Operator	Description
`$and`	Logical AND
`$or`	Logical OR
`$not`	Negates a condition
`$nor`	Opposite of OR (none must match)

CODE and OUTPUT:-

```
test> show dbs;
admin      40.00 KiB
config     48.00 KiB
local     104.00 KiB
practice  304.00 KiB

test> use test;
already on db test

test> db.createCollection("products");
{ ok: 1 }

test> db.products.insertOne({name: 'Laptop', category: 'Electronics', price: 40000});
{
  acknowledged: true,
  insertedId: ObjectId('67fdeee3aa4d1653a24aa58c')
}
```

```
test> db.products.insertMany([
...   {name: 'Smartphone', category: 'Electronics', price: 10000},
...   {name: 'Books', category: 'Stationary', price: 500}
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67fdeeedaa4d1653a24aa58d'),
    '1': ObjectId('67fdeeedaa4d1653a24aa58e')
  }
}

test> db.products.insertMany([
...   {name: 'Headphones', category: 'Tech Gadgets', price: 2500},
...   {name: 'Blender', category: 'Home Appliances', price: 3000},
...   {name: 'Notebook', category: 'Stationary', price: 50},
...   {name: 'T-shirt', category: 'Fashion', price: 800},
...   {name: 'Gaming Mouse', category: 'Tech Gadgets', price: 1500},
...   {name: 'Desk Lamp', category: 'Home Decor', price: 1200},
...   {name: 'Backpack', category: 'Accessories', price: 1800},
...   {name: 'Shoes', category: 'Fashion', price: 2500},
...   {name: 'Water Bottle', category: 'Utilities', price: 300},
...   {name: 'Tablet', category: 'Tech Gadgets', price: 20000}
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67fdef18aa4d1653a24aa58f'),
    '1': ObjectId('67fdef18aa4d1653a24aa590'),
    '2': ObjectId('67fdef18aa4d1653a24aa591'),
    '3': ObjectId('67fdef18aa4d1653a24aa592'),
    '4': ObjectId('67fdef18aa4d1653a24aa593'),
    '5': ObjectId('67fdef18aa4d1653a24aa594'),
    '6': ObjectId('67fdef18aa4d1653a24aa595'),
    '7': ObjectId('67fdef18aa4d1653a24aa596'),
    '8': ObjectId('67fdef18aa4d1653a24aa597'),
    '9': ObjectId('67fdef18aa4d1653a24aa598')
  }
}

test> db.products.find();
[
  {
    _id: ObjectId('67fdeee3aa4d1653a24aa58c'),
    name: 'Laptop',
    category: 'Electronics',
    price: 40000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58d'),
    name: 'Smartphone',
    category: 'Electronics',
    price: 10000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
    price: 500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa58f'),
    name: 'Headphones',
    category: 'Tech Gadgets',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa590'),
    name: 'Blender',
    category: 'Home Appliances',
    price: 3000
  },
```

```
  {
    _id: ObjectId('67fdef18aa4d1653a24aa591'),
    name: 'Notebook',
    category: 'Stationary',
    price: 50
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa592'),
    name: 'T-shirt',
    category: 'Fashion',
    price: 800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa593'),
    name: 'Gaming Mouse',
    category: 'Tech Gadgets',
    price: 1500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa594'),
    name: 'Desk Lamp',
    category: 'Home Decor',
    price: 1200
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa595'),
    name: 'Backpack',
    category: 'Accessories',
    price: 1800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa596'),
    name: 'Shoes',
    category: 'Fashion',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa597'),
    name: 'Water Bottle',
    category: 'Utilities',
    price: 300
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa598'),
    name: 'Tablet',
    category: 'Tech Gadgets',
    price: 20000
  }
]
```

```
test> db.products.find({name: 'Books'});
[
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
    price: 500
  }
]
```

```
test> db.products.find({price: {$lt: 20000}});
[
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58d'),
    name: 'Smartphone',
    category: 'Electronics',
    price: 10000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
```

```
    price: 500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa58f'),
    name: 'Headphones',
    category: 'Tech Gadgets',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa590'),
    name: 'Blender',
    category: 'Home Appliances',
    price: 3000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa591'),
    name: 'Notebook',
    category: 'Stationary',
    price: 50
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa592'),
    name: 'T-shirt',
    category: 'Fashion',
    price: 800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa593'),
    name: 'Gaming Mouse',
    category: 'Tech Gadgets',
    price: 1500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa594'),
    name: 'Desk Lamp',
    category: 'Home Decor',
    price: 1200
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa595'),
    name: 'Backpack',
    category: 'Accessories',
    price: 1800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa596'),
    name: 'Shoes',
    category: 'Fashion',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa597'),
    name: 'Water Bottle',
    category: 'Utilities',
    price: 300
  }
]

test> db.products.updateOne(
...   {name: 'Laptop'},
...   {$set: {price: 70000}}
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

test> db.products.find({name: 'Laptop'});
[
```

```
  {
    _id: ObjectId('67fdeee3aa4d1653a24aa58c'),
    name: 'Laptop',
    category: 'Electronics',
    price: 70000
  }
]

test> db.products.updateMany(
...   {category: 'Electronics'},
...   {$set: {category: 'Tech Gadgets'}}
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}

test> db.products.find({category: 'Tech Gadgets'});
[
  {
    _id: ObjectId('67fdeee3aa4d1653a24aa58c'),
    name: 'Laptop',
    category: 'Tech Gadgets',
    price: 70000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58d'),
    name: 'Smartphone',
    category: 'Tech Gadgets',
    price: 10000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa58f'),
    name: 'Headphones',
    category: 'Tech Gadgets',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa593'),
    name: 'Gaming Mouse',
    category: 'Tech Gadgets',
    price: 1500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa598'),
    name: 'Tablet',
    category: 'Tech Gadgets',
    price: 20000
  }
]

test> // Products in Tech Gadgets OR price < 10000
... db.products.find({
...   $or: [
...     {category: 'Tech Gadgets'},
...     {price: {$lt: 10000}}
...   ]
... });
[
  {
    _id: ObjectId('67fdeee3aa4d1653a24aa58c'),
    name: 'Laptop',
    category: 'Tech Gadgets',
    price: 70000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58d'),
    name: 'Smartphone',
    category: 'Tech Gadgets',
```

```
    price: 10000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
    price: 500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa58f'),
    name: 'Headphones',
    category: 'Tech Gadgets',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa590'),
    name: 'Blender',
    category: 'Home Appliances',
    price: 3000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa591'),
    name: 'Notebook',
    category: 'Stationary',
    price: 50
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa592'),
    name: 'T-shirt',
    category: 'Fashion',
    price: 800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa593'),
    name: 'Gaming Mouse',
    category: 'Tech Gadgets',
    price: 1500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa594'),
    name: 'Desk Lamp',
    category: 'Home Decor',
    price: 1200
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa595'),
    name: 'Backpack',
    category: 'Accessories',
    price: 1800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa596'),
    name: 'Shoes',
    category: 'Fashion',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa597'),
    name: 'Water Bottle',
    category: 'Utilities',
    price: 300
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa598'),
    name: 'Tablet',
    category: 'Tech Gadgets',
    price: 20000
  }
]
```

```
test> // Products in category "Stationary" AND price < 1000
... db.products.find({
```

```
...   $and: [
...     {category: 'Stationary'},
...     {price: {$lt: 1000}}
...   ]
... });
[
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
    price: 500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa591'),
    name: 'Notebook',
    category: 'Stationary',
    price: 50
  }
]

test> // Not Stationary
... db.products.find({
...   category: {$not: {$eq: 'Stationary'}}
... });
[
  {
    _id: ObjectId('67fdeee3aa4d1653a24aa58c'),
    name: 'Laptop',
    category: 'Tech Gadgets',
    price: 70000
  },
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58d'),
    name: 'Smartphone',
    category: 'Tech Gadgets',
    price: 10000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa58f'),
    name: 'Headphones',
    category: 'Tech Gadgets',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa590'),
    name: 'Blender',
    category: 'Home Appliances',
    price: 3000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa592'),
    name: 'T-shirt',
    category: 'Fashion',
    price: 800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa593'),
    name: 'Gaming Mouse',
    category: 'Tech Gadgets',
    price: 1500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa594'),
    name: 'Desk Lamp',
    category: 'Home Decor',
    price: 1200
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa595'),
    name: 'Backpack',
    category: 'Accessories',
    price: 1800
```

```
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa596'),
    name: 'Shoes',
    category: 'Fashion',
    price: 2500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa597'),
    name: 'Water Bottle',
    category: 'Utilities',
    price: 300
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa598'),
    name: 'Tablet',
    category: 'Tech Gadgets',
    price: 20000
  }
]

test> // Not Tech Gadgets and price < 50000
... db.products.find({
...   $nor: [
...     {category: 'Tech Gadgets'},
...     {price: {$gte: 50000}}
...   ]
... });
[
  {
    _id: ObjectId('67fdeeedaa4d1653a24aa58e'),
    name: 'Books',
    category: 'Stationary',
    price: 500
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa590'),
    name: 'Blender',
    category: 'Home Appliances',
    price: 3000
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa591'),
    name: 'Notebook',
    category: 'Stationary',
    price: 50
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa592'),
    name: 'T-shirt',
    category: 'Fashion',
    price: 800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa594'),
    name: 'Desk Lamp',
    category: 'Home Decor',
    price: 1200
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa595'),
    name: 'Backpack',
    category: 'Accessories',
    price: 1800
  },
  {
    _id: ObjectId('67fdef18aa4d1653a24aa596'),
    name: 'Shoes',
    category: 'Fashion',
    price: 2500
  },
  {
```

```
    _id: ObjectId('67fdef18aa4d1653a24aa597'),
    name: 'Water Bottle',
    category: 'Utilities',
    price: 300
  }
]

test> db.products.deleteOne({name: 'Laptop'});
{ acknowledged: true, deletedCount: 1 }

test> db.products.deleteMany({ category: 'Fashion' });
{ acknowledged: true, deletedCount: 2 }
```

CONCLUSION:-

In conclusion, this assignment demonstrates the effective use of MongoDB for performing CRUD (Create, Read, Update, Delete) operations on a product database. It included creating collections, inserting documents, retrieving specific records using filters and logical operators, updating fields, and deleting data conditionally. The assignment highlights MongoDB's flexibility in handling document-based data and showcases its efficiency for managing real-world product inventories in a NoSQL environment.

ASSIGNMENT:- 5

AIM:- Data Visualization

PROBLEM STATEMENT:- Connect to data, Build Charts and Analyze Data, Create Dashboard, Create Stories using Tableau/PowerBI.

THOERY:-

Data Visualization is a critical step in the data analysis process, allowing users to interpret data through interactive and visual representations. Tools like Power BI and Tableau are widely used in industry for connecting to various data sources, building meaningful visualizations, creating dashboards, and telling interactive data stories that drive business insights and decision-making.

This assignment involves working with a real-world dataset titled *"Data Professionals Survey"*, and demonstrating the complete workflow in both Power BI or Tableau — from data connection and transformation to interactive dashboards and story-building.

Objectives

- Connect to and clean data using Tableau or Power BI.
- Create various types of charts and visualizations.
- Analyze trends and derive insights.
- Design dashboards to present KPIs and metrics.
- Use storytelling features to guide end-users through insights interactively.

Dataset Description

The dataset data_professionals_survey.csv contains responses from data professionals around the world, covering aspects such as:

- Age and Salary
- Job Titles
- Country of residence
- Preferred programming languages
- Happiness with salary and work
- Perceived difficulty of entering the data profession

This dataset helps us understand demographics, salary distribution, tool preferences, and satisfaction levels among data professionals.

Implementation Steps

1. **Connect to Data**
   - **Power BI:-** Used "Get Data" → CSV → Loaded the dataset and transformed data using Power Query Editor for cleaning (e.g., renaming columns, handling null values).
   - **Tableau:-** Used "Text File" connection to load the dataset. Pivoted programming language columns for unified analysis and checked data types for consistency.

2. Build Charts and Analyze Data

Different charts were created to explore the dataset:

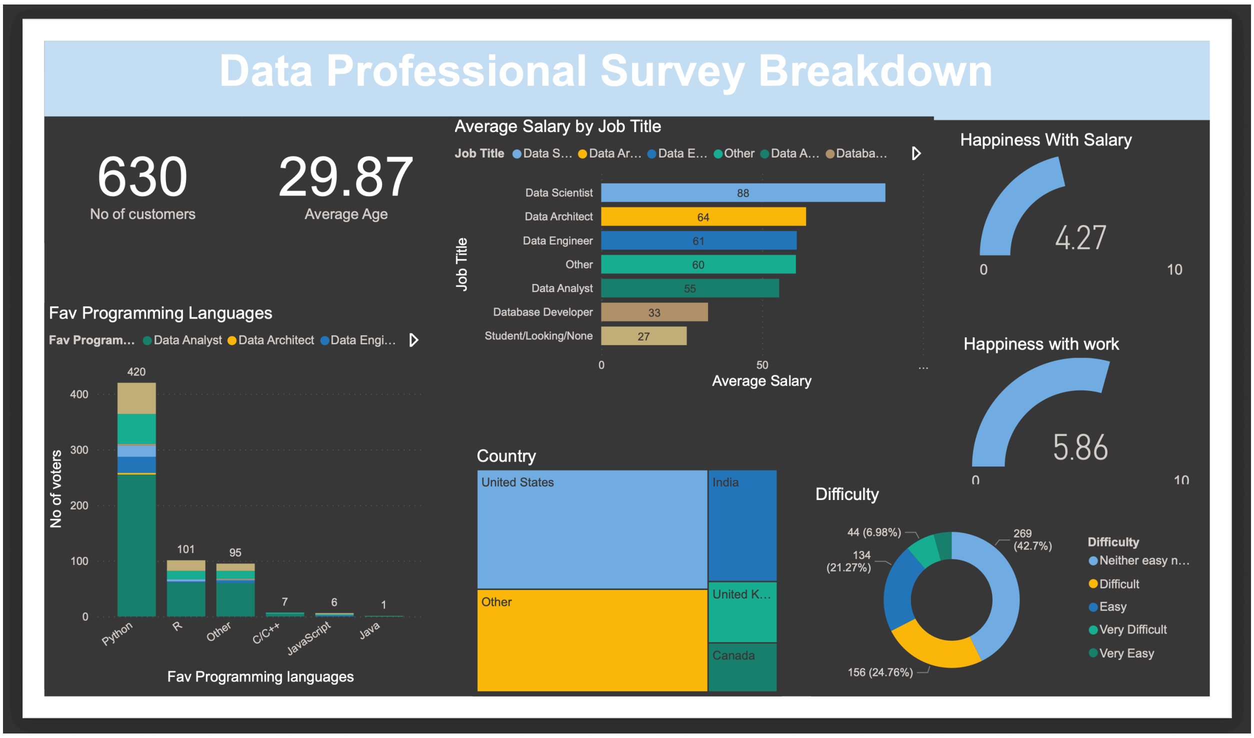
- **Bar Charts:-** To analyze job title distribution and most preferred programming languages.
- **Line Charts:-** To visualize trends like salary across age groups.
- **Pie/Donut Charts:-** To show the country-wise distribution.
- **KPI Cards:-** To display average age, total respondents, and average salary.
- **Table/Matrix Views:-** For detailed job title and salary breakdown.
- **Maps (in Tableau):-** For geographic distribution of participants.

DAX formulas in Power BI and Calculated Fields in Tableau were used to derive additional insights like high-earning roles, average happiness scores, etc.

3. Create Dashboards

Dashboards were built by logically arranging the visuals with consistent themes:

- Overview Dashboard: KPIs, job title chart, and country distribution.
- Salary Insights: Salary by job title and satisfaction scores.
- Technical Preferences: Programming languages and tool choices.
- Filters like Country, Job Title, and Age were added for interactivity.



CONCLUSION:-

This assignment demonstrated how Tableau and Power BI can be used to uncover valuable insights from a dataset through visual storytelling. It highlighted the importance of effective data cleaning, appropriate visualization selection, interactive design, and storytelling in deriving and communicating insights. Both tools empowered us to explore data in depth and share findings in a meaningful and visually engaging way.